SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTD215
Application of C++ in Multimedia

Tutor-Marked Assignment

July 2025 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 15% of the final mark for MTD215 Application of C++ in Multimedia.

The cut-off date for this assignment is **6 October 2025, 23:55 hrs**.

Note to Students:

You are to include the following particulars in your submission: Course Code, Title of the TMA, SUSS PI No., Your Name, and Submission Date.

Question 1: Team Assignment Optimization (20 marks)

Design and explain a C++ class called Circle that encapsulates the properties of a circle and supports basic geometric calculations. The program should follow object-oriented programming principles and include clear comments explaining each part.

(a) Define the Circle class:

i. Two private member variables: radius and pi (set pi = 3.1416)

ii. Constructor that takes radius as input and initializes the object.

iii. Member functions to compute:

- Area of the circle
- Circumference
- Diameter

(10 marks)

(b) Write and explain a main() function that

i. Prompts the user to input a radius value.

ii. Creates a Circle object with the input.

iii. Displays the calculated area, circumference, and diameter. (6 marks)

(c) Extend the program using SDL or SFML to visually draw the circle based on the radius entered by the user.

(4 marks)

Question 2: Team Assignment Optimization (25 marks)

You are organizing a talent show, and each of 12 participants is scored (1–10) based on their audition. Write and explain a C++ program to assign them into three groups of four, such that the total scores of each group are as balanced as possible.

(a) Accept the names and scores of 12 participants via user input.

(8 marks)

(b) Implement an algorithm to form 3 groups with minimal difference in total group scores. Display the participants grouped accordingly.

(10 marks)

(c) Implement and explain an algorithm to form 3 groups with minimal difference in total group scores. Display the participants grouped accordingly. Print the total score of each group and the maximum difference between any two groups.

(7 marks)

Question 3: 1D Puzzle Game (25 marks)

Design and implement a simple console-based puzzle game in C++ where the player is given a 1D array of 10 unique random numbers. The goal is to sort the numbers by repeatedly swapping two adjacent numbers.

(a) Generate and explain 10 unique random numbers and display them in a horizontal line using ASCII formatting.

(5 marks)

(b) Allow the user to input two indices to attempt a swap. Reject illegal (non-adjacent) moves with an appropriate error message.

(5 marks)

(c) Perform the swap if valid and re-display the updated array. Repeat until sorted.

(5 marks)

(d) When the array is fully sorted, congratulate the player and end the game.

(5 marks)

(e) Modularize the code: write separate functions for:

- Generating the numbers
- Displaying the array
- Checking adjacency
- Verifying if sorted

(5 marks)

Question 4: SDL Animation and Comparison (30 marks)

(a) Using SDL, design and apply a simple animation where a colored square moves left to right and bounces back upon reaching the edge of the window.

- Create a window of size 800×600.
- Animate a square of size 50×50 pixels.
- Add a "Pause/Resume" toggle button to control the animation.

(18 marks)

(b) Compare SDL with OpenGL and SFML for multimedia/graphics applications:

Sl #	Library/API	Pros	Cons
1	SDL		
2	OpenGL		
3	SFML		

Briefly explain and discuss which one you would choose for a beginner-friendly multimedia application and why.

(12 marks)

---- END OF ASSIGNMENT ----